

Lijie Ding

Contact: (401)-410-4049, ljding.jobs@gmail.com
For more: [Github](#), [LinkedIn](#), [Google Scholar](#), [CV](#)

Skills

AI/ML: Agentic AI, Large Language Models (LLMs), RAG, Physics-informed Machine Learning, Deep Learning, Bayesian inference, CNNs, VAEs, Gaussian Processes

AI Frameworks: PyTorch, CrewAI, Agent SDK, scikit-learn, Claude Code, Codex

Scientific Computing: Monte Carlo (MC), Molecular Dynamics (MD), LAMMPS, HOOMD-blue, HPC, Slurm

Programming: Python, C/C++, NumPy, Pandas, SQL, Git

Experience

Oak Ridge National Laboratory (ORNL)

Postdoctoral Research Associate

Oak Ridge, TN

May 2024 – Present

Agentic AI for Science: Engineered multi-agent systems (CrewAI, Agent SDK, and Agent Skills) and applied RAG to orchestrate complex scientific workflows such as SAS data analysis, MD (LAMMPS), and MC (HOOMD-blue) simulations.

LLM Application: Architected an LLM-driven proposal review pipeline using pairwise preference and AI-as-a-judge, achieving 800× cost reduction while maintaining review performance for Spallation Neutron Source.

AI for Characterization: Developed and trained physics-informed ML/DL models in PyTorch (e.g., Gaussian process regression, Bayesian inference, VAEs, CNNs) to extract hidden information from SAS, applied to colloids, polymers, lamellae, and polydisperse systems.

Scientific Computing: Developed MD/MC simulations on HPC clusters (Slurm) using C++, Python, LAMMPS, and HOOMD-blue for colloids, polymers, active fluid and rheology systems.

Goldman Sachs

Quantitative Strategist, Vice President (Jan24-May24), Associate (Jun22-Dec 23)

New York, NY

Jun 2022 – May 2024

Model Development: Owned the full lifecycle development of proprietary production valuation models, ensuring high reliability and performance for interest rate (IR) derivative pricing. Executed the SOFR transition for IR products and conducted statistical validation for SOFR-based pricing.

Large-Scale Attribution Analysis: Built analysis pipelines to explain price variance vs. market consensus and decompose moves into key risk factors for multi-billion-dollar inventory instruments using Python and SQL.

Brown University

Ph.D. Research Assistant

Providence, RI

Sep 2017 – Jun 2022

Scientific Computing: Developed high-performance MC simulations for colloidal membranes in C++. Performed mathematical modeling and numerical studies in Python.

Scientific Data Analysis: Developed OpenCV-based image analysis tools for DNA Brownian motion and performed time series analysis.

Publications

Lijie Ding et al., “ToPolyAgent: AI Agents for Coarse-Grained Bead-Spring Topological Polymer Simulations,” *Digital Discovery* **5**, 901–909 (2026). DOI: [10.1039/D5DD00471C](https://doi.org/10.1039/D5DD00471C)

Lijie Ding et al., “LLMs Can Assist with Proposal Selection at Large User Facilities,” (2025). arXiv: [2512.10895](https://arxiv.org/abs/2512.10895)

Lijie Ding et al., “Machine Learning-Assisted Profiling of a Kinked Ladder Polymer Structure using Scattering,” *Digital Discovery* **4**, 1570–1577 (2025). DOI: [10.1039/D5DD00051C](https://doi.org/10.1039/D5DD00051C)

H. Du, J. Su, J. Li, **Lijie Ding**, et al., “Which LLM Multi-Agent Protocol to Choose?,” *Proceedings of the International Conference on Machine Learning (ICML)* (2026). OpenReview: [GZRiXE2zBe](https://openreview.net/forum?id=GZRiXE2zBe)

More: full list can be found at [Google Scholar](#), or [CV](#)

Education

Brown University

Ph.D. in Physics (Dissertation: Chiral Liquid Crystals on Deformable Surfaces)

Providence, RI

2022

University of Science and Technology of China (USTC)

B.Sc. in Applied Physics

Hefei, China

2017